

Kathmandu Engineering College
Department of Electronics, Communication and Information



Introduction to Signals, Time Division Multiplexing, and Delta Modulation

Lab Report 3

Submitted by:

Buddha Shrestha

Student ID: KAT080BEI06

Semester: III/II

Submitted to:

**Department of Electronics,
Communication and Information**

June 23, 2026

1 Objective

- To implement Time Division Multiplexing (TDM) and Demultiplexing techniques for signal mixing and recovery.
- To model and evaluate a Delta Modulation (DM) loop using configurable step sizes.

2 Theory

2.1 Pulse Amplitude Modulation (PAM)

The amplitude of regularly spaced rectangular pulses (i.e., a carrier pulse train) is varied according to the instantaneous value of the modulating message signal. The output signal is in the form of pulses modulated to trace the profile of the incoming analog signal while adhering to Nyquist criteria bounds. A primary real-world application of PAM signals is found within **Time Division Multiplexing (TDM)** lines.

2.1.1 Drawbacks of Pulse Amplitude Modulation (PAM)

- The channel bandwidth required for the transmission of a PAM signal is exceptionally large compared to the highest frequency present in the baseband modulating signal.
- Because the absolute amplitudes of individual carrier pulses vary directly with the message envelope, noise interference impacts the system at maximum levels and cannot be easily isolated or removed.
- Since pulse amplitude maps directly to instantaneous power, amplitude fluctuations vary the peak power demands required at the transmitter terminal.

2.2 Time Division Multiplexing (TDM)

Time Division Multiplexing is a method of transmitting multiple independent data streams over a single communication channel by dividing the total available transmission timeline into non-overlapping, distinct time slots. This interleaves sample components sequentially so that the full bandwidth of the physical medium is dynamically shared among separate low-rate sources.

Periodic sampling leaves predictable structural intervals between successive tracking pulses. Consequently, the empty duration between adjacent pulses of a given PAM wave can be allocated to other independent message channels on a time-shared basis. This allows a common channel link to stack several simultaneous user files without risking signal overlap or multi-channel interference.

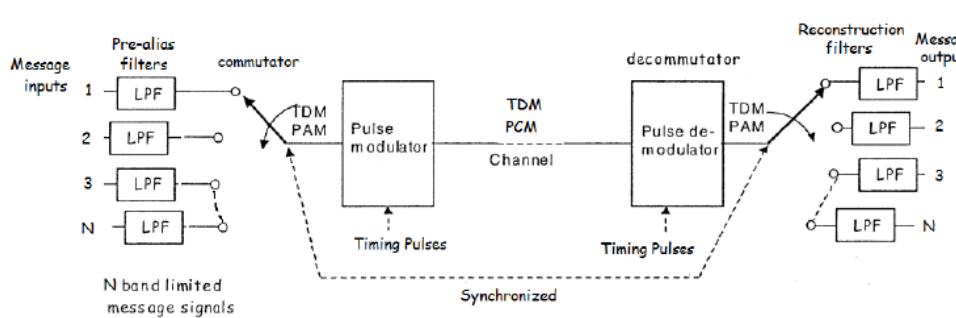


Figure 1: System block diagram layout of a Time Division Multiplexing framework.

The baseline architecture of a TDM pipeline is illustrated in Figure 1. Initially, each input message path passes through a low-pass pre-alias filter to eliminate high frequencies that could

otherwise distort data boundaries during sampling. The output paths match up at an electronic commutator switch. The systematic operations of this commutator block are configured:

- To extract a narrow sample from each of the N input channels at a precise sampling rate $f_s > 2f_m$, where f_m represents the upper cutoff boundary of the pre-alias filter.
- To interleave these N extracted elements sequentially inside a single unified frame allocation interval defined by $T_s = 1/f_s$.

This multiplexed collection feeds into a pulse-amplitude modulator block to condition the stream for propagation across the shared medium. Let T_s denote the full sampling period of an individual user path, and let T_x signify the compressed time spacing separating adjacent slot points in the resulting composite multiplexed train. The relational constraint maps explicitly as:

$$T_x = \frac{T_s}{N} \quad (1)$$

Thus, squeezing N independent channels into a single transmission slot introduces a spectrum bandwidth expansion factor of N . At the destination terminal, the composite wave enters a pulse amplitude demodulator which undoes the original framing layout. A specialized de-commutator, running in rigid synchronization with the transmitter switch, then distributes these split pulses into independent low-pass reconstruction filters to output the original target messages.

2.3 Delta Modulation (DM)

Exploiting adjacent sample correlations within communication systems implies that oversampling a baseband message signal (sampling at rates far exceeding the Nyquist minimum limit) will significantly increase compliance between consecutive structural samples. **Delta Modulation (DM)** acts as a highly streamlined, single-bit (two-level) quantization subset of Differential Pulse Code Modulation (DPCM) designed around this principle.

In its functional operation, a DM system generates a rough staircase tracking envelope to approximate the incoming oversampled analog waveform. The variance between the raw input and this local staircase tracking approximation is checked by a single-bit comparator, producing two representation tracking states: $+\delta$ and $-\delta$. If the current staircase approximation trails beneath the input signal level at a sampling epoch, it steps up by an increment of δ . If it sits above the input, it steps down by δ . So long as the source signal does not experience high-frequency changes that exceed the tracking rate, the tracking step remains within safe bounds of the real wave.

Representing the analog input signal as $m(t)$ and the quantized staircase tracking trace as $m_q(t)$, the foundational mathematical steps of Delta Modulation are characterized through these discrete relations:

$$e(nT_s) = m(nT_s) - m_q(nT_s - T_s) \quad (2)$$

$$e_q(nT_s) = \delta \cdot \text{sgn}[e(nT_s)] \quad (3)$$

$$m_q(nT_s) = m_q(nT_s - T_s) + e_q(nT_s) \quad (4)$$

where T_s is the system sampling period, $e(nT_s)$ denotes the error difference tracking the gap between the incoming sample $m(nT_s)$ and the previous internal approximation $m_q(nT_s - T_s)$, and $e_q(nT_s)$ provides the bit stream output indicating positive or negative trends.

The primary asset of Delta Modulation lies in its structural simplicity. The processing loop relies on a summer, a basic threshold comparator, and an accumulator block. Solving the system tracking relations yields the absolute accumulator state at any instant nT_s :

$$m_q(nT_s) = \delta \sum_{i=1}^n \text{sgn}[e(iT_s)] = \sum_{i=1}^n e_q(iT_s) \quad (5)$$

At each clock tick, the accumulator changes the tracked output by a value of δ . If the input signal amplitude values exceed the historical internal reference step, a positive increment $(+\delta)$ is applied. If it falls below, a negative step $(-\delta)$ is registered, allowing the processing hardware to trace the targeted wave one step at a time.

3 Source Code & Simulation Output Layouts

3.1 Original Input Signals Generation

```
clc;
clear all;
close all;

% Define tracking domain indices
x=0:0.5:4*pi;

% Channel 1: Sinusoidal source
signal1=8*sin(x);
l=length(signal1);

% Channel 2: Triangular source
signal2=8*triang(l)';

figure(1);
subplot(2,2,1);
plot(signal1);
title('Sine Wave-Buddha Shrestha');
xlabel('time'); ylabel('Amplitude');

subplot(2,2,2);
stem(signal1);
title('Discrete Sine Wave-Buddha Shrestha');
xlabel('n'); ylabel('X[n]');

subplot(2,2,3);
plot(signal2);
title('Triangular Wave-Buddha Shrestha');
xlabel('Time'); ylabel('Amplitude');

subplot(2,2,4);
stem(signal2);
title('Discrete Triangular Wave-Buddha Shrestha');
xlabel('n'); ylabel('X[n]');
```

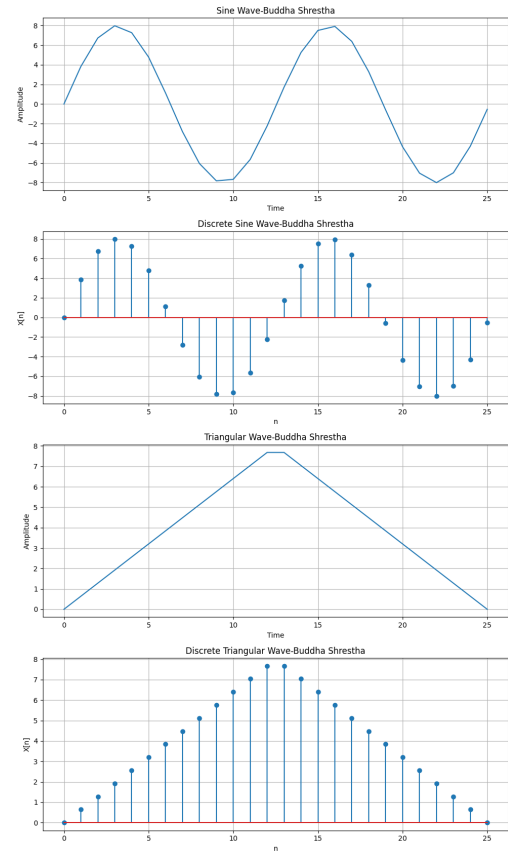


Figure 2: Primary source channels visual plots.

3.2 Time Division Multiplexing (TDM) Execution

```
% Interleave source matrices sequentially
l1=length(signal1);
signal = zeros(2, l1);

for i=1:l1
    signal(1,i)=signal1(i);
    signal(2,i)=signal2(i);
end

figure(2);
tdm_signal=reshape(signal,1,2*l1);
subplot(1,1,1);
stem(tdm_signal);
title('TDM Multiplication in discrete form
      - Buddha Shrestha');
xlabel('n'); ylabel('X[n]');
```

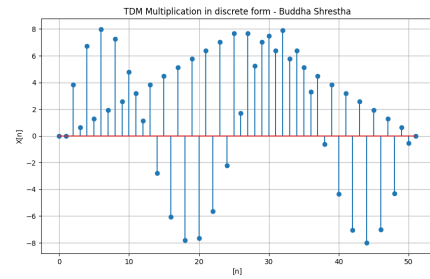


Figure 3: Interleaved composite TDM stream plot.

3.3 Demultiplexing & Signal Reconstruction

```
% Reshape composite stream back to dual
array rows
demux=reshape(tdm_signal,2,l1);
signal3 = zeros(1, l1);
signal4 = zeros(1, l1);

for i=1:l1
    signal3(i)=demux(1,i);
    signal4(i)=demux(2,i);
end

figure(3);
subplot(2,1,1);
plot(signal3);
title('recovered sinusoidal signal -
      Buddha Shrestha');
xlabel('time'); ylabel('Amplitude');

subplot(2,1,2);
plot(signal4);
title('Recovered triangle signal - Buddha
      Shrestha');
xlabel('Time'); ylabel('Amplitude');
```

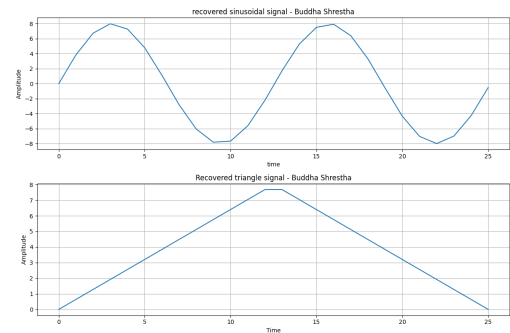


Figure 4: Demultiplexed destination channels.

3.4 Delta Modulation (DM) System Simulation

```
% Delta Modulation Loop Architecture
del = 0.4; % Step size parameter
definition
t=0:2*pi/100:2*pi;
x=5*sin(t);

figure(4);
plot(x, 'LineWidth', 1.5);
hold on;

y=[0];
xr=zeros(1, length(x));

for i=1:length(x)-1
    if xr(i) < x(i)
        d=1;
        xr(i+1)=xr(i)+del;
    elseif xr(i)==x(i)
        d=1;
        xr(i+1)=xr(i);
    else
        d=0;
        xr(i+1)=xr(i)-del;
    end
    y=[y d];
end
stairs(xr, 'r', 'LineWidth', 1.2);
title('Delta Modulation Simulation for
    del_step = 0.5 - Buddha Shrestha');
xlabel('Samples'); ylabel('Amplitude');
legend('Input Source', 'DM Approximation')
;
grid on;
hold off;

mse=sum((x-xr).^2)/length(x);
```

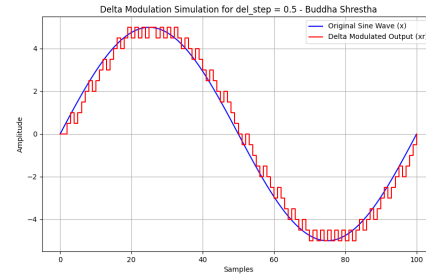


Figure 5: Delta Modulation approximation tracking.

4 Discussion

This experiment demonstrates how computational environments like MATLAB can be utilized to model core signaling types and transmission architectures:

- **Time Continuity Mapping:** Setting up continuous-time signals and transforming them into sampled discrete domains via localized tracking stems showcases the foundational theorem parameters of discrete mapping.
- **TDM Channel Multiplexing:** Interleaving independent functional arrays (Sinusoidal and Triangular) using high-speed loop restructuring demonstrated zero cross-talk communication when clock cycles are perfectly matched. Reversing the allocation via `reshape()` properties verified clean retrieval without channel bleed.
- **Delta Modulation Bounds:** Simulating a discrete tracking loop verified that the precision of tracking a continuous input profile depends directly on step size configuration (Δ). Setting step scales incorrectly causes slope overload distortion or granular noise artifacts.
- In both **Pulse code modulation** and **differential pulse code modulation**, the pulse-repetitive frequency or the sampling rate is chosen to be slightly greater than the **Nyquist rate** (i.e., greater than twice the highest frequency component) of the analog signal. In **Delta modulation**, on the other hand, the sampling rate is purposely chosen to be much higher than the Nyquist rate. The reason for such a choice in the latter case is to increase the Nyquist rate. The reason for such a choice in the latter case is to increase correlation between adjacent samples derived from the information-bearing analog signal and thereby to simplify the physical implementation of the delta modulation process.

5 Conclusion

The experimental investigation into foundational signals, Time Division Multiplexing mechanics, and Delta Modulation quantization setups has been successfully completed using MATLAB.

The practical outputs confirmed that:

- Discrete transformations cleanly inherit structural properties of continuous waveforms when sampled above critical limits.
- Multi-channel communication pipelines can safely combine independent streams by utilizing orthogonal time slot mapping layouts.
- Differential quantization models like Delta Modulation can successfully reconstruct source tracking envelopes using an intuitive single-bit encoding architecture.